

NSCOM03 Midterm Cheat Sheet — Modules 1–4

L01–L04 · one lecture per page · grey = slide page · bold = key term · colors follow the unit: bps blue, baud orange, Hz green, levels violet, power/SNR/dB magenta, seconds amber

L01 Review of the Physical & Data Link Layers

L01 · Layers & data link

- OSI bottom-up: **Physical, Data link, Network, Transport, Session, Presentation, Application**
- Slide TCP/IP (4): Application = OSI 5–7 · Transport = 4 · **Internet = 3** · **Network access = 1–2**
- Slide 2 examples: switch, bridge, Ethernet, PPP → L2; hubs, repeaters, coax, fiber → L1; IP → L3; TCP/UDP → L4
- DLL: frames moved **error-free between two end nodes** over the physical layer
- DLL services: **framing**, frame sequencing, flow control, error detection (sometimes correction), **QoS**
- Physical layer: frames → electrical/optical/electromagnetic signals; cables, connectors, pins

L01 · Networks & signals

- LAN typically within **5 km**, copper/fiber, IEEE 802 · WAN: fiber, submarine cable, satellite & terrestrial microwave
- **Dark fiber** = unused fiber · Starlink **550 km**, 775 satellites (Oct 2020)
- Wireless: unguided, cells; Bluetooth, 802.11, LTE
- IEEE 802: .2 LLC · **.3 Ethernet (CSMA/CD)** · .4 token bus · .5 token ring · .11 wireless · .13 none · .16 WLL
- MAC styles: **ALOHA style** vs **token style**
- Analog: vary amplitude/frequency/phase · Digital: e.g. 1 = +5 V, 0 = 0 V
- Analog transmission when **bandwidth is limited** → modulator / demodulator

L02 Signals, Bandwidth, Noise & Data-Rate Limits

Frequency and period

L02 p6; L02 p7

$$f = \frac{1}{T} \quad T = \frac{1}{f}$$

f frequency — cycles per second (Hz)

T period — the time one cycle takes (s)

Switch between how often a signal repeats and how long one cycle lasts. The prefixes flip: ms ↔ kHz, μs ↔ MHz.

Slide example: Slide 7: 4 cycles in 1 s → $f = 4$ Hz and $T = \frac{1}{4}$ s.

Wavelength

L02 p9

$$\lambda = \frac{c}{f} = c \times T$$

λ wavelength — distance covered during one period (m)

c propagation speed (3×10^8 m/s for light in free space) (m/s)

f frequency (Hz)

T period (s)

How far the signal travels while it completes one cycle — it depends on the medium (through c) and on the frequency.

Slide example: Slide 10: red light, $f = 4 \times 10^{14}$ Hz →

$$\lambda = \frac{3 \times 10^8}{4 \times 10^{14}} = 0.75 \mu\text{m}.$$

Bandwidth

L02 p13

$$B = f_{\text{high}} - f_{\text{low}}$$

B bandwidth — width of the range of frequencies (Hz)

f_{high} highest frequency in the composite signal (Hz)

f_{low} lowest frequency in the composite signal (Hz)

The span of frequencies a composite signal contains (or a medium passes) — not the highest frequency.

Slide example: Slide 13: components from 1000 to 5000 Hz → $B = 5000 - 1000 = 4000$ Hz.

Bits per level

L02 p14

$$\text{bits per level} = \log_2 L$$

L number of signal levels

A digital signal with L levels carries $\log_2 L$ bits in each level: 2 levels → 1 bit, 4 → 2, 8 → 3.

Slide example: Slide 14: $L = 4 \rightarrow \log_2 4 = 2$ bits per level; 8 levels in 1 s → $N = 8 \times 2 = 16$ bps.

Decibel

L02 p21; L02 p22

$$\text{dB} = 10 \log_{10} \frac{P_2}{P_1}$$

dB gain (+) or loss (−) between two points (dB)

P_1 power at the first point (W)

P_2 power at the second point (W)

Compare two powers. Negative = attenuated, positive = amplified; the dB values of a chain of stages simply add.

Slide example: Slide 22: power halved → $10 \log_{10} 0.5 = 10 \times (-0.3) = -3$ dB.

Signal-to-noise ratio

L02 p25

$$\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} \quad \text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$$

SNR signal-to-noise ratio, a plain ratio

P_{signal} average signal power (W)

P_{noise} average noise power (W)

SNR_{dB} the same ratio in decibels (dB)

How far the signal stands above the noise. Shannon needs the plain ratio — convert from dB first.

Slide example: Slide 26: $\text{SNR} = \frac{10,000 \mu\text{W}}{1 \mu\text{W}} = 10,000 \rightarrow 10 \log_{10} 10^4 = 40$ dB.

Nyquist bit rate (noiseless)

L02 p29

$$N = 2 \times B \times \log_2 L$$

N maximum bit rate of a noiseless channel (bps)

B channel bandwidth (Hz)

L number of signal levels

Noiseless channel: the most bits per second a bandwidth B can carry with L levels. More levels raise N , but the receiver must still tell them apart.

Slide example: Slide 30: $N = 2 \times 3000 \times \log_2 2 = 6000$ bps.

Shannon capacity (noisy)

L02 p31

$$C = B \times \log_2(1 + \text{SNR})$$

C capacity — the upper limit on the bit rate (bps)

B channel bandwidth (Hz)

SNR signal-to-noise ratio as a plain ratio (not dB)

Noisy channel: the highest possible bit rate, whatever the signalling. There is no L in it — use Shannon for the limit, then Nyquist to choose L .

Slide example: Slide 32:

$$C = 3000 \times \log_2(1 + 3162) \approx 3000 \times 11.62 = 34,860 \text{ bps.}$$

Latency

L02 p35

$$\text{Latency} = T_{\text{prop}} + T_{\text{trans}} + T_{\text{queue}} + T_{\text{proc}}$$

T_{prop} propagation time = distance $\div$ propagation speed (s)

T_{trans} transmission time = message size (bits) $\div$ bandwidth (bps) (s)

T_{queue} waiting time in device queues (s)

T_{proc} processing time in devices (s)

Total time for a whole message to arrive. Long distances make propagation dominate; large messages on slow links make transmission dominate.

Slide example: Slide 37: 12,000 km at 2.4×10^8 m/s $\rightarrow$

$T_{\text{prop}} = 50$ ms; 2500 bytes at 1 Gbps $\rightarrow T_{\text{trans}} = 0.020$ ms.

L02 · Signals & bandwidth

- Analog = infinite levels · digital = limited levels; periodic vs nonperiodic
- $f = 1/T$ (Hz) · 4 cycles in 1 s $\rightarrow 4$ Hz, $T = 1/4$ s · phase $0^\circ/90^\circ/180^\circ/270^\circ = 0\uparrow$ / peak / $0\downarrow$ / trough
- $\lambda = c/f$, $c = 3 \times 10^8$ m/s · red light 4×10^{14} Hz $\rightarrow 0.75 \mu\text{m}$
- Composite = sum of sines (Fourier); $B = f_{\text{high}} - f_{\text{low}}$ (5000 – 1000 = 4000 Hz)
- Digital signal = composite with **infinite bandwidth**; L levels $\rightarrow \log_2 L$ bits per level
- Baseband $\rightarrow$ low-pass channel (first harmonic $B = N/2$) · broadband $\rightarrow$ modulation onto a **bandpass** channel

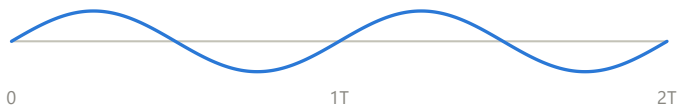
L02 · dB, SNR & noise

- **dB** = $10 \log_{10}(P_2/P_1)$; loss negative, gain positive; half power = **−3 dB**
- dB of cascaded stages add; $P_{\text{out}} = P_{\text{in}} \times 10^{(\text{dB}/10)}$
- **SNR** = P_s/P_n ; $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR} \cdot 10 \text{ mW} / 1 \mu\text{W} \rightarrow 10,000 \rightarrow 40 \text{ dB}$
- Noise: **thermal** (electrons), **induced** (motors), **crosstalk** (adjacent wire), **impulse** (spikes)
- Attenuation = energy loss · distortion = shape change

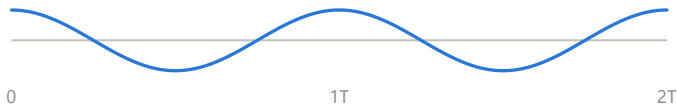
L02 · Rate limits & latency

- Nyquist (noiseless): $N = 2B \log_2 L \cdot 3000 \text{ Hz}$, $L = 2 \rightarrow$ **6000 bps**
- Shannon (noisy): $C = B \log_2(1 + \text{SNR})$, SNR linear · 3000 Hz, 3162 $\rightarrow \approx$ **34.86 kbps**
- Slide 33: 1 MHz, SNR 63 $\rightarrow C =$ **6 Mbps**; use 4 Mbps $\rightarrow L = 4$
- **Latency** = $T_{\text{prop}} + T_{\text{trans}} + T_{\text{queue}} + T_{\text{proc}}$
- $T_{\text{prop}} = d/v$ · $T_{\text{trans}} = \text{size}/\text{bandwidth}$ · 12,000 km at $2.4 \times 10^8 \text{ m/s} \rightarrow$ **50 ms**; 2.5 KB at 1 Gbps $\rightarrow$ **0.020 ms**
- Throughput < bandwidth · jitter = variation in delay

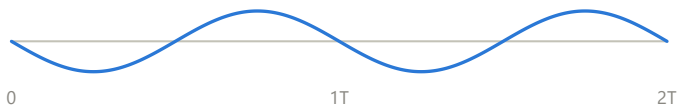
0° — starts at 0 going up



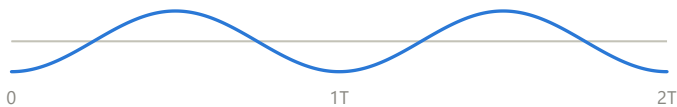
90° — starts at the peak



180° — starts at 0 going down



270° — starts at the trough



Phase: the same sine wave shifted by 0°, 90°, 180° and 270° (slide 8).

L03a Digital Transmission I: Line Coding, Block Coding & Scrambling

Signal (baud) rate

L03 p8

$$S = c \times N \times \frac{1}{r}$$

S signal rate — signal elements per second (baud)

c case factor: worst 1, best 0, average $\frac{1}{2}$

N data rate (bps)

r data elements carried by one signal element

How many signal elements per second a line code needs for data rate N . Use the average case $c = \frac{1}{2}$ unless the question says otherwise.

Slide example: Slide 9: $S = \frac{1}{2} \times 100,000 \times \frac{1}{1} = 50,000$ baud
= 50 kbaud.

Minimum bandwidth

L03 p25; L03 p46

$$B_{\min} = S_{\text{avg}}$$

$B_{\min}$ minimum bandwidth the line code needs (Hz)

S_{avg} average signal rate (baud)

Per scheme (slide 46): NRZ-L, NRZ-I, AMI $\rightarrow N/2 \cdot RZ$,
Manchester, differential Manchester $\rightarrow N \cdot 2B1Q \rightarrow N/4 \cdot \text{MLT-3} \rightarrow N/3 \cdot 8B6T \rightarrow 3N/4 \cdot 4D\text{-PAM5} \rightarrow N/8$.

Slide example: Slide 25: NRZ-I at $N = 1$ Mbps $\rightarrow S = 500$ kbaud $\rightarrow B_{\min} = 500$ kHz.

mBnL code condition

L03 p36; L03 p37

$$2^m \leq L^n$$

m data bits in a block (data elements)

n signal elements per block

L levels per signal element: B = 2, T = 3, Q = 4

An mBnL code works only if the L^n signal patterns are at least as many as the 2^m data patterns; the patterns left over are redundancy.

Slide example: Slides 36–37: 8B6T $\rightarrow 2^8 = 256 \leq 3^6 = 729$
 $\rightarrow$ valid, 473 patterns unused.

4B/5B rate

L03 p54

$$N_{\text{line}} = N \times \frac{5}{4}$$

N_{line} bit rate on the line after 4B/5B (bps)

N data rate before block coding (bps)

Every 4 data bits become 5 code bits, so the line carries 25% more bits; then apply the line code to the new rate.

Slide example: Slide 54: $N = 1$ Mbps $\rightarrow N_{\text{line}} = 1.25$ Mbps;
with NRZ-I $B_{\min} = 625$ kHz.

L03a · Rates & properties

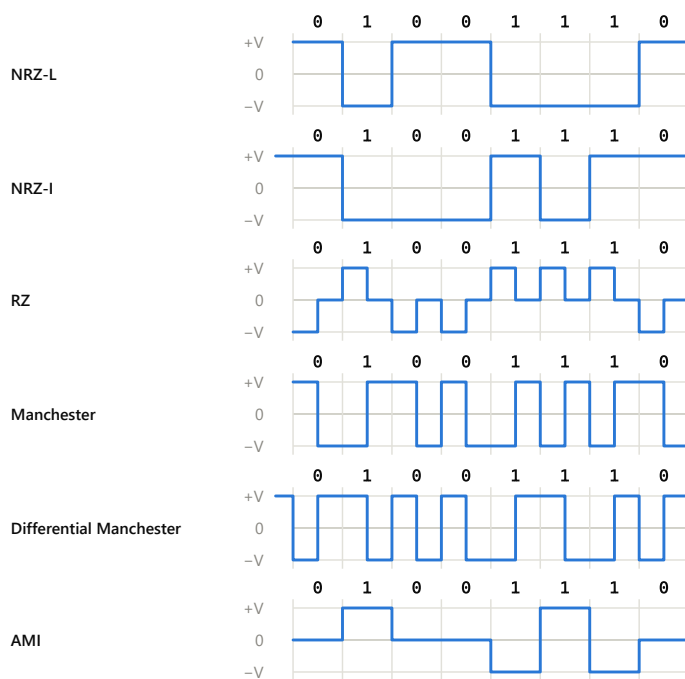
- $S = c \times N \times 1/r$ (baud); $c = \frac{1}{2}$ average; $B_{\min} = S$
- r : NRZ 1 · RZ/Manchester $\frac{1}{2}$ · 2B1Q 2 · 8B6T 4/3
- Avg B: NRZ-L/I, AMI $N/2$ · RZ, biphase $N \cdot 2B1Q$ $N/4$ · MLT-3 $N/3$ · 8B6T $3N/4$ · 4D-PAM5 $N/8$
- Good code: no **baseline wandering**, no **DC**, self-sync, error detection, noise immunity, low complexity
- Ex 1: 100 kbps, $r = 1 \rightarrow 50$ kbaud · Ex 2: 0.1% fast $\rightarrow 1001$ bps / 1,001,000 bps · Ex 3: NRZ-I 1 Mbps $\rightarrow 500$ kbaud, 500 kHz

L03a · Scheme rules (slide figures)

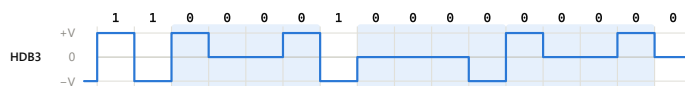
- Unipolar NRZ 1 = +V, 0 = 0 · **NRZ-L** 0 = +V, 1 = -V · **NRZ-I: 1 inverts** (start +)
- RZ: 1 = +V $\rightarrow$ 0, 0 = -V $\rightarrow$ 0 · **Manchester** 0 = high $\rightarrow$ low, 1 = low $\rightarrow$ high
- **Diff. Manchester: 0 = transition at start**, always mid-bit transition
- **AMI** 0 = 0 V, 1s alternate (first +) · pseudoternary = reverse (low priority)
- 2B1Q (prev +): 00 +1, 01 +3, 10 -1, 11 -3 (prev -: signs flip)
- MLT-3: 0 stays; 1 $\rightarrow$ 0 from $\pm V$, or $\rightarrow$ opposite of last nonzero from 0
- mBnL needs $2^m \leq L^n$; B = 2, T = 3, Q = 4 levels

L03a · Block coding & scrambling

- Block coding mB/nB: division → substitution → combination; adds redundancy
- **4B/5B + NRZ-I**; rate $\times 5/4$; 16 spare codes (control: I 11111, Q 00000, H 00100, J 11000, K 10001, T 01101)
- Ex 4: 1 Mbps → **1.25 Mbps** → NRZ-I **625 kHz** / Manchester 1.25 MHz
- 8B/10B = 5B/6B + 3B/4B + disparity controller
- **B8ZS: 8 zeros** → **000VB0VB** (V same polarity as previous pulse)
- **HDB3: 4 zeros** → **B00V if even, 000V if odd** (pulses since last substitution)

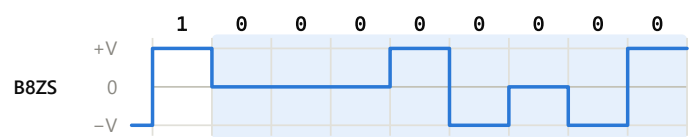


01001110 in the slide conventions (slides 23, 27, 29, 32).



Pulse before the stream: -. Substitutions: bits 3–6 → B00V (even pulses since last substitution); bits 8–11 → 000V (odd pulses since last substitution); bits 12–15 → B00V (even pulses since last substitution). Marks: B at bit 3, V at bit 6, V at bit 11, B at bit 12, V at bit 15.

HDB3: 000V after an odd count, B00V after an even count (slide 61).



Pulse before the stream: -. Substitutions: bits 2–9 → 000VB0VB. Marks: V at bit 5, B at bit 6, V at bit 8, B at bit 9. B8ZS: 000VB0VB (slide 59).

Sampling rate

L03 p64

$$f_s = \frac{1}{T_s}$$

 f_s sampling rate (samples/s) T_s sampling interval (s)

Samples per second from the time between samples, and back.

Slide example: Telephone voice: $f_s = 8000$ samples/s $\rightarrow$
 $T_s = 125 \mu\text{s}$.

Nyquist sampling theorem

L03 p66; L03 p67

$$f_s \geq 2 \times f_{\max}$$

 f_s sampling rate (samples/s) $f_{\max}$ highest frequency in the signal (Hz)

Sample at least twice the highest frequency — for low-pass and bandpass signals alike. A bandpass signal's bandwidth alone is not enough: you need its highest frequency.

Slide example: Slide 69: voice up to 4 kHz $\rightarrow$
 $f_s = 2 \times 4000 = 8000$ samples/s.

Zone height (quantization step)

L03 p72

$$\Delta = \frac{V_{\max} - V_{\min}}{L}$$

 Δ zone height (quantization step) (V) $V_{\max}$ top of the amplitude range (V) $V_{\min}$ bottom of the amplitude range (V) L number of zones (levels)

Split the amplitude range into L equal zones; each sample is replaced by the midpoint of its zone.

Slide example: Slide 74: -20 V to $+20$ V with $L = 8 \rightarrow$
 $\Delta = \frac{40}{8} = 5$ V.

Bits per sample

L03 p75

$$n_b = \log_2 L$$

 n_b bits per sample L number of zones (levels)

The number of bits needed to give every zone its own code.

Slide example: $L = 8$ zones $\rightarrow n_b = 3$ bits per sample (slide 76 codes 000 to 111).

PCM bit rate

L03 p79

$$N = n_b \times f_s$$

 N PCM bit rate (bps) n_b bits per sample f_s sampling rate (samples/s)

The bit rate PCM produces: bits per sample $\times$ samples per second.

Slide example: Slide 80: $N = 8 \times 8000 = 64,000$ bps = 64 kbps.

Minimum PCM bandwidth

L03 p83

$$B_{\min} = n_b \times f_{\max} = \frac{N}{2}$$

 $B_{\min}$ minimum bandwidth of the PCM signal (Hz) n_b bits per sample $f_{\max}$ highest frequency of the analog signal (Hz) N PCM bit rate (bps)

Minimum bandwidth when the PCM bits are sent with an NRZ-type line code ($c = 1/2$, $r = 1$).

Slide example: Slide 83: $B_{\min} = 8 \times 4 \text{ kHz} = 32 \text{ kHz}$.
 Slide example 10: $8 \times 4 \text{ kHz} = 32 \text{ kHz}$.

Quantization error bound

L03 p76; L03 p78

$$|\text{error}| \leq \frac{\Delta}{2}$$

Δ zone height (V)

The midpoint of a zone is never more than half a zone away from the true sample.

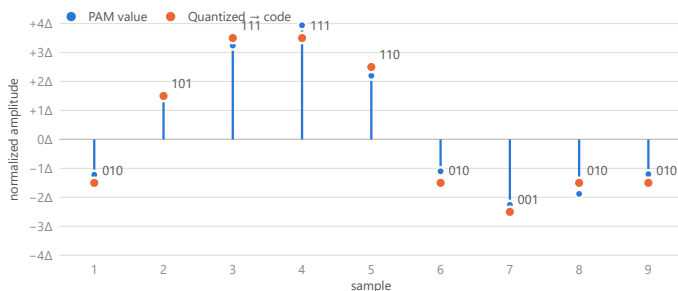
Slide example: Slide 76: $\Delta = 5$ V, so the error is at most 2.5 V; -6.1 V becomes -7.5 V, an error of -1.4 V.

L03b · PCM

- PCM = **sample** → **quantize** → **encode**; filter first
- $f_s = 1/T_s$; Nyquist $f_s \geq 2 \cdot f_{\max}$ (low-pass and bandpass); voice 4 kHz → **8000 sps**
- Bandpass with only B known → rate **cannot be determined**
- Sampling: ideal (impulse) · natural (follows curve) · **flat-top** (sample and hold) → PAM
- $\Delta = (\max - \min)/L$; zones 0 ... L-1 from the bottom; sample → **midpoint**; $n_b = \log_2 L$
- $-20 \dots +20$ V, $L = 8 \rightarrow \Delta = 5$ V; 000 = -20 to -15 V ... 111 = $+15$ to $+20$ V
- $|\text{error}| \leq \Delta/2$; low amplitudes suffer → non-linear quantization, **companding**

L03b · Rates, DM, modes

- $N = n_b \times f_s$ — voice, 8 bits → **64 kbps**
- $B_{\min} = n_b \times f_{\max}$ (NRZ-type) — 8×4 kHz = **32 kHz**
- Decoder: **hold circuit** + low-pass filter (cutoff = $f_{\max}$)
- DM: 1 bit per sample, 1 = up δ , 0 = down δ ; DPCM = quantized difference
- Parallel: n bits per tick on n wires · Serial: 1 bit per tick
- Async: **start 0** + byte + **stop 1**, gaps OK · Sync: no start/stop, frames · Isochronous: equal gaps



Quantization: zones, midpoints and 3-bit codes (slide 76).

L04 Digital-to-Analog: ASK, FSK, PSK & QAM

Baud (signal) rate

L04 p5

$$S = N \times \frac{1}{r}$$

S baud rate — signal elements per second (baud)

N bit rate (bps)

r data bits carried by one signal element

Analog transmission of digital data: each signal element carries r bits, so $S \leq N$.

Slide example: Slide 6: 1000 baud $\times$ 4 bits per element $\rightarrow N = 4000$ bps.

Analog transmission of digital data: $S \leq N$.

Bits per element $\leftrightarrow$ levels

L04 p7

$$r = \log_2 L \quad \Leftrightarrow \quad L = 2^r$$

r bits per signal element

L number of distinct signal elements (levels, phases, frequencies)

Bits per signal element from the number of distinct elements, and back.

Slide example: Slide 7: 8000 bps at 1000 baud $\rightarrow r = 8 \rightarrow L = 2^8 = 256$.

ASK / PSK bandwidth

L04 p9; L04 p21

$$B = (1 + d) \times S$$

B bandwidth (Hz)

d modulation and filtering factor, $0 \leq d \leq 1$

S baud rate (baud)

Bandwidth of ASK and of PSK. With $d = 0$ the bandwidth is at its minimum, $B = S$.

Slide example: Slide 12: a 100 kHz band with $d = 1 \rightarrow S = \frac{100}{2} = 50$ kbaud $\rightarrow N = 50$ kbps.

Binary FSK bandwidth

L04 p14; L04 p15

$$B = (1 + d) \times S + 2\Delta f$$

B bandwidth (Hz)

d modulation and filtering factor, $0 \leq d \leq 1$

S baud rate (baud)

Δf half the carrier spacing: bit 1 $\rightarrow f_c + \Delta f$, bit 0 $\rightarrow f_c - \Delta f$ (Hz)

Binary FSK needs the ASK bandwidth plus the spacing $2\Delta f$ between its two carriers.

Slide example: Slide 16: 100 kHz band, $2\Delta f = 50$ kHz, $d = 1 \rightarrow 2S = 50$ kHz $\rightarrow S = 25$ kbaud, $N = 25$ kbps.

Multilevel FSK bandwidth

L04 p18; L04 p19

$$B = (1 + d) \times S + (L - 1) \times 2\Delta f$$

B bandwidth (Hz)

S baud rate (baud)

L number of frequencies

Δf half the spacing between neighbouring frequencies (Hz)

L frequencies, each $2\Delta f$ apart. With $d = 0$ and the minimum spacing $2\Delta f = S$, this becomes $B = L \times S$.

Slide example: Slide 19: 3 Mbps, 3 bits at a time $\rightarrow L = 8$, $S = 1$ Mbaud, $B = 8 \times 1 = 8$ MHz.

Δ Slide 18 prints " $(L - 1)/2\Delta f$ " — it is a product.

Carrier in an available band

L04 p12; L04 p13

$$f_c = \frac{f_{\text{low}} + f_{\text{high}}}{2}$$

f_c carrier frequency (Hz)

f_{low} bottom of the available band (Hz)

f_{high} top of the available band (Hz)

Put the carrier in the middle of the band. Full duplex: split the band into two halves with a carrier in the middle of each.

Slide example: Slide 12: 200–300 kHz $\rightarrow f_c = 250$ kHz; full duplex (slide 13): 225 kHz and 275 kHz.

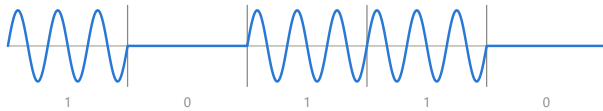
L04 · Digital → analog

- **ASK** amplitude · **FSK** frequency · **PSK** phase · **QAM** amplitude + phase
- $S = N/r$ (baud); $r = \log_2 L$, $L = 2^r$; $S \leq N$
- ASK/PSK: $B = (1+d)S$, $0 \leq d \leq 1$; BASK = OOK, $r = 1$
- FSK: $B = (1+d)S + 2\Delta f$; bit 1 $\rightarrow f_c + \Delta f$, bit 0 $\rightarrow f_c - \Delta f$
- MFSK: $B = (1+d)S + (L-1) \cdot 2\Delta f = L \cdot S$ when $d = 0$, $2\Delta f = S$
- Carrier at band middle; full duplex $\rightarrow$ halve the band (200–300 kHz $\rightarrow$ 225 and 275 kHz)
- BASK = **unipolar** NRZ $\times$ carrier; BPSK = **polar** NRZ $\times$ carrier; FSK via VCO
- PSK more robust than ASK (noise hits amplitude)

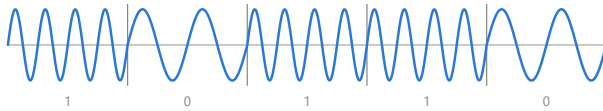
L04 · Constellations & examples

- $x = I$ (in-phase), $y = Q$ (quadrature); length = amplitude, angle = phase
- OOK: 0 at origin, 1 on $+I$ · BPSK: 0 at $-I$, 1 at $+I$
- QPSK (1st bit $\rightarrow I$): **11 $\rightarrow 45^\circ$, 01 $\rightarrow 135^\circ$, 00 $\rightarrow -135^\circ$, 10 $\rightarrow -45^\circ$**
- 16-QAM (slide 31): first 2 bits = quadrant (00 I, 10 II, 11 III, 01 IV); 1100 $\rightarrow 225^\circ$
- $r = 4$, $S = 1000 \rightarrow$ **4000 bps** · $N = 8000$, $S = 1000 \rightarrow$ **$r = 8$, $L = 256$**
- ASK 200–300 kHz, $d = 1 \rightarrow$ **250 kHz, 50 kbps** · FSK, $2\Delta f = 50$ kHz $\rightarrow$ **25 kbps**
- QPSK 12 Mbps, $d = 0 \rightarrow$ **6 MHz** · MFSK 3 bits, 3 Mbps $\rightarrow$ **$L = 8$, 1 Mbaud, 8 MHz**

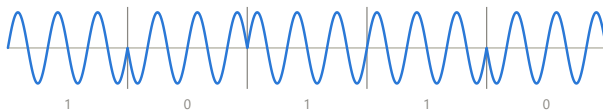
ASK (OOK)



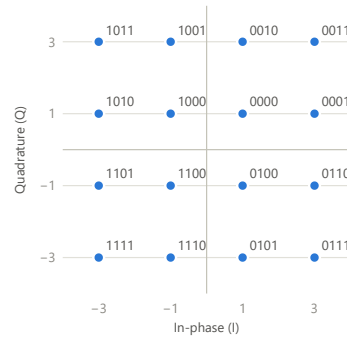
Binary FSK



BPSK



10110 in ASK, FSK and BPSK (slides 10, 14, 22).



16-QAM constellation: amplitude = distance, phase = angle (slide 31).

Σ Exam conventions, mental math and traps

Symbol clashes

- **S = signal (baud) rate** in baud: $S = c \cdot N/r$ (L03), $S = N/r$ (L04). **C = Shannon capacity** in bps. Not the same thing.
- **c = case factor** (0 to 1, average $\frac{1}{2}$) in $S = c \cdot N/r$. In L02, **c = speed of light**, 3×10^8 m/s.
- **N = bit (data) rate** in bps. $N_{\max} = 2 \cdot B \cdot \log_2 L$ is Nyquist's limit for a noiseless channel.

Unit traps

- **K = 1000** for data rates and sizes (1 kbps = 1000 bps; 1 KB = 1000 B on the slides), not 1024.
- Bytes $\times 8$ = bits (2.5 KB = 20,000 bits) before dividing by a bit rate.
- **$\mu\text{s} = 10^{-6}$ s, ms = 10^{-3} s**: 0.020 ms = 20 μs .
- Convert everything to base units (Hz, bps, s, m) before substituting.

Mental-math tips (no calculator)

- $\log_2$ of a power of 2 = **count the doublings** ($\log_2 64 = 6$); $\log_2 1000 \approx 10$
- dB $\rightarrow$ ratio: **$10^{(\text{dB}/10)}$** ; ratio $\rightarrow$ dB: $10 \cdot \log_{10}(\text{ratio})$. Add dB to multiply ratios.
- **3 dB $\approx \times 2$, 10 dB $\approx \times 10$** : 20 dB = $\times 100$, -3 dB $\approx \times \frac{1}{2}$
- **$2^{10} \approx 10^3$** (1024 vs 1000)
- Powers of 2: $2^1=2 \cdot 2^2=4 \cdot 2^3=8 \cdot 2^4=16 \cdot 2^5=32 \cdot 2^6=64 \cdot 2^7=128 \cdot 2^8=256 \cdot 2^9=512 \cdot 2^{10}=1024$
- SNR = $2^k - 1$ gives $\log_2(1 + \text{SNR}) = k$: SNR 3, 7, 15, 31, 63, 255 $\rightarrow$ 2, 3, 4, 5, 6, 8

END OF DOCUMENT: cheatsheet · build r1